From Censorship to Detoxification: Rewriting Harmful Memes

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Abstract

Harmful memes often rely on image-text interactions to imply identity-based attacks. We study meme text detoxification: given a meme image and caption, generate a safer caption that preserves topic and image-text coherence. Our system uses LLaVA-NeXT as a multimodal teacher to generate structured explanations and pseudo-rewrites, then distills this behavior into a LoRA-adapted BART-large student. We also test a CLIP Proxy + BART inference path, where CLIP image/text features are mapped to BART encoder soft tokens, avoiding LLaVA/DetoxLLM-sized generation at deployment. On 280 held-out memes, the full BART student reaches 0.963 ± 0.006 Text STA and 0.207 ± 0.006 STA delta across five seeds, numerically exceeding a single-run text-only DetoxLLM-7B baseline on automatic toxicity metrics. The proxy obtains the highest BERTScore and CLIPScore but lower Text STA, indicating a fidelity-safety trade-off rather than stronger safety.

1. Introduction

Hateful meme moderation is often framed as classification: predict whether an image-text pair is harmful and route it for review or removal [1, 2]. This is necessary but incomplete. Meme captions can be ambiguous alone, while the image supplies the target, stereotype, or implied insult. A pure text detoxifier can miss that interaction; a pure detector does not construct a less harmful alternative that preserves topic and context.

We formulate harmful meme moderation as *text rewriting*. Given a meme image and overlaid text, the goal is to rewrite the text so that the harmful framing is reduced while the topic, short meme style, and visual compatibility remain recognizable. This raises two questions. First, can multimodal teacher explanations supervise a compact meme rewriter? Second, can CLIP-derived soft encoder states provide useful visual grounding without running LLaVA/DetoxLLM-sized generation?

Our contributions are threefold. First, we implement an explain-then-rewrite pipeline that uses LLaVA-NeXT [3] to produce target groups, visual evidence, implicit meanings, and pseudo-detoxified captions. Second, we fine-tune BART-large [4] with LoRA [5] on these meme-specific pseudo-labels and compare it to DetoxLLM-7B [6] and non-fine-tuned BART. Third, we propose a deployment-oriented CLIP Proxy + BART path: CLIP [7] encodes the image and text, a small MLP predicts BART encoder-memory soft tokens, and the fine-tuned BART decoder generates the rewrite without running LLaVA or a DetoxLLM-sized causal LM at inference.

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2. Related Work

Hateful meme understanding. The Hateful Memes benchmark showed that multimodal reasoning is required because unimodal cues are deliberately confounded [1]. VisualBERT [2] and CLIP [7] provide useful image-text representations, but the standard objective is still detection. Our work differs by producing a revised caption rather than only a moderation label.

Text detoxification. ParaDetox casts detoxification as parallel style transfer for toxic sentences [8], and DetoxLLM trains large causal models for detoxification with explanations [6]. These methods are primarily text-only at inference. We use them as the closest external task baseline, but our supervision is meme-specific and grounded in teacher-generated multimodal explanations.

Multimodal teachers and efficient students. LLaVA-NeXT improves visual instruction following and OCR-oriented reasoning [3], which makes it suitable for teacher pseudo-labeling. However, using a 7B VLM at deployment is costly. We therefore distill the teacher into BART-large and test whether CLIP-derived soft encoder states can preserve useful visual context. Because evaluation relies on automatic fidelity/alignment metrics, we report BERTScore [9] and CLIPScore [10] as proxies rather than human judgments. This leaves a gap between detecting harmful multimodal implications and generating a safer, visually coherent alternative; we explore it with multimodal teacher supervision and a CLIP-conditioned deployment path.

3. Method

Data construction. We combine HarMeme [11], MAMI [12], and MMHS150K [13]. Stage 0 extracts text with EasyOCR and keeps images with 10–300 OCR characters. CLIP then filters non-meme images: HarMeme and MAMI use a binary meme-vs.-screenshot comparison, while MMHS150K uses a five-prompt rule and keeps an image only if the meme prompt is top and has probability at least 0.45. Filtered examples are split 80/10/10, stratified by dataset and harmful label.

Teacher pseudo-labels. For each split, LLaVA-NeXT first generates an explanation with three fields: target group, visual evidence, and implicit harmful meaning. A second prompt asks for a short non-hateful rewrite that keeps the meme topic and style. Stage 1 records all attempts but marks pass/fail using Text STA, BERTScore, and toxicity-drop filters. The Stage 2 builder keeps rows that passed Stage 1, have no parse error, show positive toxicity reduction, satisfy sanitized length, artifact, repetition, symbol, and no-edit checks, and have normalized source/rewrite similarity at most 0.95. The final supervised data contains 3,578 train, 269 validation, and 280 held-out test examples.

BART student. The student starts from BART-large and is trained on the meme pseudo-rewrite data. Encoder inputs contain the original text plus optional teacher fields in an explicit detoxification instruction. We train four conditions: `full`, `target_only`, `visual_only`, and `none`. LoRA is applied to BART attention projections and feed-forward layers with rank 32, alpha 64, dropout 0.05, learning rate 10^{-4}, batch size 8, warm-up 50 steps, weight decay 0.01, and 5 epochs. About 17M of roughly 400M parameters are trainable.

CLIP Proxy + BART. The proxy is a 3-layer MLP. It receives concatenated CLIP image and text embeddings (1536 dimensions) and predicts a sequence of $K = 16$ BART encoder soft tokens of size 1024. During training, the target is the adaptive-average-pooled encoder memory of the full-condition fine-tuned BART. At inference, proxy tokens are concatenated with BART encoder states from a text-only null-context prompt, then decoded by the fine-tuned BART decoder. The architecture is illustrated in Figure 1. This turns multimodal grounding into a continuous prefix for the decoder rather than requiring a textual explanation from a VLM at inference. It is our main deployment path because it keeps visual input through CLIP while removing the LLaVA teacher and DetoxLLM-sized causal LM from inference.

4. Data Licensing and Ethics

The project uses HarMeme [11], MAMI [12], and MMHS150K [13] only under the original providers' access terms. We do not redistribute images, captions, or derived

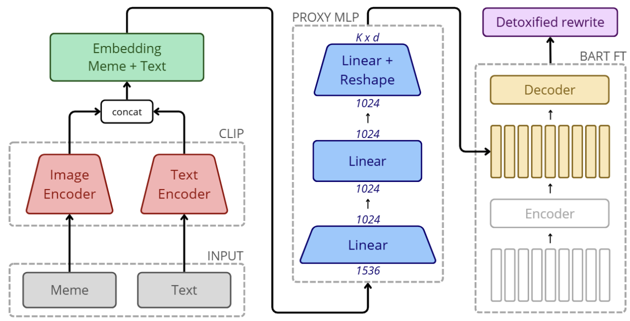

Figure 1. Efficient inference path. Frozen CLIP encodes image/text; a trained MLP predicts BART soft tokens for the fine-tuned decoder, avoiding LLaVA/DetoxLLM-sized inference.

harmful examples; the report contains only aggregate metrics, and access notes are documented in the repository. HarMeme is downloaded by the pipeline, MAMI is manual-request, and MMHS150K uses the authors' research links.

Ethically, detoxification should be read as a harm-reduction tool, not as an automatic moderation authority. Rewriting can remove slurs while still preserving stereotypes, or it can over-neutralize legitimate counterspeech and satire. Toxicity classifiers and hate datasets also encode annotation, platform, privacy, and consent biases for social-media-derived images. Deployment would require human review, uncertainty reporting, and clear separation between educational research and real moderation decisions.

5. Experiments and Results

All systems are evaluated on the same 280 held-out examples with the same automatic scoring pipeline and exact outputs: no manual correction or post-hoc sanitization is applied. Reference systems include the fixed LLaVA pseudo-label teacher and a single-run text-only DetoxLLM-7B baseline. Metrics are Text STA (mean non-toxic probability from a RoBERTa classifier), STA delta (rewrite STA minus original STA), BERTScore F1, and normalized CLIPScore. Higher is preferable only jointly: Text STA and STA delta estimate toxicity reduction, while BERTScore/CLIPScore are fidelity/alignment proxies and may reward copying. BART-FT is averaged across five LoRA seeds; LLaVA and DetoxLLM are single runs. Proxy variance reflects the same five BART students, not independent proxy seeds.

Across five BART seeds, all BART-FT variants obtain numerically higher mean Text STA and STA delta than the single-run text-only DetoxLLM baseline. The `full` condition has the strongest BART-only automatic toxicity metrics, while `target_only` is close and has the best BART-only SIM. The ablation suggests teacher-field conditioning gives little measurable gain under these automatic toxicity metrics. The LLaVA teacher row is a pseudo-label reference, not an

Model	Text STA	STA Δ	SIM	CLIP
LLaVA teacher	**0.990**	**0.234**	0.472	0.636
DetoxLLM-7B	0.940	0.184	0.443	0.634
BART-FT full	0.963±0.006	0.207±0.006	0.330±0.013	0.628±0.001
BART-FT target	0.960±0.005	0.204±0.005	0.335±0.013	0.629±0.002
BART-FT visual	0.960±0.004	0.204±0.004	0.284±0.080	0.624±0.007
BART-FT none	0.957±0.013	0.201±0.013	0.325±0.030	0.628±0.002
Proxy+BART	0.884±0.015	0.128±0.015	**0.581±0.020**	**0.638±0.002**

Table 1. Test results on 280 memes. SIM=BERTScore F1; CLIP=normalized CLIPScore. ± is mean±sample std over five BART seeds.

independent baseline.

Base BART often generated generic, off-task safe text rather than meaningful rewrites, supporting the need for meme-specific fine-tuning. DetoxLLM keeps higher SIM than the BART-only students (0.443 vs. roughly 0.33), but copying can inflate BERTScore when toxic input is retained. Similarity must therefore be interpreted with Text STA and STA delta, not as a stand-alone quality score.

The proxy obtains the best mean SIM (0.581) and CLIPScore (0.638), suggesting stronger source retention and image alignment, but Text STA drops to 0.884. This may reflect harmful-content retention, so the proxy is an efficient deployment trade-off, not the strongest detoxifier. Its advantage is architectural: CLIP plus a small MLP and BART decoder, not LLaVA. In our A100 benchmark, BART-FT requires 122 ms per rewrite, versus 9.38 s for LLaVA and 1.99 s for DetoxLLM. Proxy evaluation took 3.4 minutes for 280 examples (0.73 s/image), roughly 13× faster than LLaVA and 2.7× faster than DetoxLLM.

Limitations. The project depends on LLaVA pseudo-labels rather than human detoxification references, and the heterogeneous source datasets remain noisy even after filtering. We do not include human evaluation; automatic toxicity, BERTScore, and CLIPScore may miss pragmatic harm, counterspeech, satire, or cases where lexical similarity preserves the harmful implication. The BART student without the proxy receives visual context only as teacher text fields, not pixels. Finally, the proxy is not a drop-in replacement: its lower Text STA requires stronger toxicity control.

6. Conclusion

We reframed harmful meme moderation as constructive text rewriting. A LLaVA-NeXT teacher supplies multimodal explanations and pseudo-rewrites, while a BART-large student gives the strongest automatic detoxification results among our BART variants and numerically exceeds the single-run text-only DetoxLLM baseline on toxicity metrics. The proposed CLIP Proxy + BART path removes LLaVA/DetoxLLM-sized inference and improves fidelity metrics, but its lower Text STA shows that efficient visual grounding must be paired with stronger toxicity control.

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